Chapter 4: Linear Dynamics

T. K. Deskins, Ph.D.

Professor of Physics & Chemistry

Southern Union State Community College

PHY201

Lecture 4

Part I

Forces and Newton's Laws of Motion

Outline of Part I: Forces and Newton's Laws of Motion

Kinematics vs. Dynamics

Kinematics (Ch. 2–3)

Describes **how** objects move.

- Position, velocity, acceleration
- Does not ask *why*

Dynamics (Ch. 4)

Explains **why** objects move.

- Forces cause changes in motion
- Asks: *what force produces this* $\vec{a}$*?*

Cause and Effect

Effect		**Cause**
Rain	$\leftarrow$	Storm clouds
Sunburn	$\leftarrow$	The Sun
Acceleration	$\leftarrow$	Net Force

Dynamics: the study of how forces affect the motion of objects.

Force

Force $\left(\vec{F}\right)$ — SI Unit: Newton (N)

A **force** is a push or a pull on an object.

Force is a **vector** quantity — it has both magnitude and direction.

Examples of Forces

- Pushing a door open
- Pulling a rope
- Gravity pulling you toward Earth
- A surface pushing back on your feet

Multiple forces on one object add to give a single **net force**:

$$\vec{F}_{\text{net}} = \sum \vec{F}$$

Because force is a vector, forces combine by **vector addition**.

Outline of Part I: Forces and Newton's Laws of Motion

Newton's First Law of Motion

Newton's First Law — The Law of Inertia

A body at rest remains at rest, or, if in motion, remains in motion at a **constant velocity** unless acted on by a **net external force**.

Inertia

Inertia is the tendency of an object to resist changes in its state of motion.

- An object at rest **stays at rest** unless pushed or pulled.
- A moving object **keeps moving** at the same speed and direction unless a net force acts.
- **Constant velocity (including rest) is the natural state.**

The First Law reframes the question:

Instead of *"why do moving objects stop?"*

ask *"what force is causing the deceleration?"*

Mass vs. Weight

Mass (m) — SI Unit: kg

- Quantity of matter in an object
- Measure of **inertia**
- **Invariant**: does not change with location

Weight ($\vec{w}$) — SI Unit: N

- Gravitational force on an object
- **Depends on location** (different on Earth vs. Moon)
- Directed **downward** toward Earth's center

Mass and weight are **not** the same thing.
Mass is measured in kg; weight is measured in N.

A 1 kg object weighs $w = mg = (1\,\mathrm{kg})(9.80\,\frac{\mathrm{m}}{\mathrm{s}^2}) = 9.80\,\mathrm{N}$ on Earth, but only $1.63\,\mathrm{N}$ on the Moon.

Check Your Understanding

Which has more mass — a kilogram of cotton balls or a kilogram of gold?

Come to lecture to see the answer.

Outline of Part I: Forces and Newton's Laws of Motion

Newton's Second Law of Motion

Newton's Second Law

The acceleration of a system is **directly proportional to** and in the **same direction** as the net external force, and **inversely proportional** to its mass:

$$\vec{F}_{\text{net}} = m\vec{a}$$

where $\vec{F}_{\text{net}}$ is the net external force (N), m is the mass (kg), and $\vec{a}$ is the acceleration $\left(\frac{\text{m}}{\text{s}^2}\right)$.

Solved for acceleration:

$$\vec{a}_x = \frac{1}{m}\vec{F}_{\text{net},x}$$

- $\vec{a} \propto \vec{F}_{\text{net}}$: double force $\Rightarrow$ double acceleration
- $\vec{a} \propto \frac{1}{m}$: double mass $\Rightarrow$ half acceleration
- $\vec{a}$ is in the **same direction** as $\vec{F}_{\text{net}}$

Net Force $\left(\vec{F}_{\text{net}}\right)$

The **net force** is the **vector sum** of all external forces acting on an object:

$$\vec{F}_{\text{net},x} = \sum \vec{F}_x$$

Analogy: Net vs. Gross

- You earn \$10/hr *(gross)*, taxes take \$1/hr, you receive \$9/hr *(net)*

Likewise: $\vec{F}_{\text{push}} = 10\,\text{N}$ and friction $\vec{f} = -1\,\text{N}$, so:

$$\vec{F}_{\text{net}} = 10\,\text{N} + (-1\,\text{N}) = \boxed{9\,\text{N}}$$

Only **external** forces contribute to $\vec{F}_{\text{net}}$. Internal forces between parts of a system cancel.

The Newton (Unit)

Definition of the Newton (N)

$$1\,\mathrm{N} = 1\,\mathrm{kg}\cdot\mathrm{m/s^2}$$

One newton is the force required to give a 1 kg object an acceleration of $1\,\frac{\mathrm{m}}{\mathrm{s}^2}$.

- A small apple weighs approximately 1 N
- A 150-lb person weighs approximately 667 N
- $1\,\mathrm{N} \approx 0.225\,\mathrm{lb}$

$$\vec{F}_{\mathrm{net},x} = m\vec{a}_x \qquad \Longleftrightarrow \qquad \mathrm{N} = \mathrm{kg}\cdot\frac{\mathrm{m}}{\mathrm{s}^2}$$

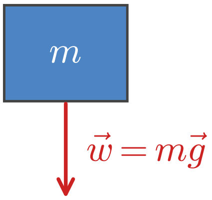
Weight

Weight ($\vec{w}$) — SI Unit: N

Weight is the **gravitational force** on an object of mass m:

$$\vec{w} = m\vec{g}$$

where m is the object's mass and $g = 9.80\,\frac{\text{m}}{\text{s}^2}$ is the magnitude of the gravitational acceleration at Earth's surface. Weight is directed **downward**.



Weight is a **force** (vector, in N).
Mass is a **scalar** (in kg).

Example: Force on a Crate

Jimmy the Monkey Pushes a Crate — (cp2e_ch4_ex3)

Jimmy the monkey pushes a 10.0 kg crate with a force of 100 N.
The friction force acting on the crate is 15.0 N.
Find the magnitude of the crate's acceleration.

$$m = 10.0\,\mathrm{kg},\ \vec{F}_p = 100\,\mathrm{N},\ \vec{f} = -15.0\,\mathrm{N}$$

$$\vec{F}_{\mathrm{net},x} = \sum \vec{F}_x = \vec{F}_p + \vec{f}$$

$$\vec{F}_{\mathrm{net},x} = 100\,\mathrm{N} + (-15.0\,\mathrm{N}) = 85.0\,\mathrm{N}$$

$$\vec{a}_x = \frac{\vec{F}_{\mathrm{net},x}}{m} = \frac{85.0\,\mathrm{N}}{10.0\,\mathrm{kg}}$$

$$\boxed{\vec{a}_x = 8.50\,\frac{\mathrm{m}}{\mathrm{s}^2}}$$

Friction opposes the direction of motion, so it is **negative** in our coordinate system.

- Always define positive direction first.
- Forces opposing motion are negative.

Check Your Understanding

Which statement is correct?

(a) Net force causes **motion**.

(b) Net force causes **change in motion**.

Explain your answer and give an example.

Come to lecture to see the answer.

Check Your Understanding

A rock is thrown straight up.

What is the net external force acting on the rock at the very **top** of its trajectory?

Come to lecture to see the answer.

Outline of Part I: Forces and Newton's Laws of Motion

Newton's Third Law of Motion

Newton's Third Law

Whenever one body exerts a force on a second body, the first body experiences a force that is **equal in magnitude** and **opposite in direction**.

Action–Reaction Pairs

Action force	Reaction force
Swimmer pushes wall **backward**	Wall pushes swimmer **forward**
Foot pushes ground **backward**	Ground pushes foot **forward**
Rocket expels gas **backward**	Gas pushes rocket **forward**

Thrust: the reaction force that propels rockets, planes, and cars forward.

Newton's Third Law — Common Misconception

Equal-and-opposite forces **do not cancel**.
They act on **different objects** — they are never part of the same free-body diagram.

- Earth pulls **you** downward (your weight).
- You pull **Earth** upward with equal force.
- These forces act on **different objects**, so they do not cancel.
- To analyze *your* motion, you use only the forces on *you*.

When analyzing a system, define the system boundary first. Forces crossing the boundary are **external**; forces entirely within the system are **internal** and cancel.

Check Your Understanding

When you take off in a jet aircraft, there is a sensation of being pushed back into your seat.

Is there actually a force pushing **backward** on you? Explain using Newton's Third Law.

Come to lecture to see the answer.

Part II

Free-Body Diagrams and Types of Forces

Outline of Part II: Free-Body Diagrams and Types of Forces

Free-Body Diagrams

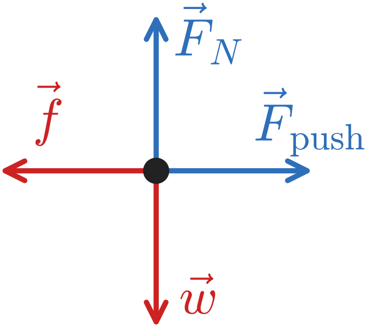
Free-Body Diagram (FBD)

A **free-body diagram** is a sketch in which:

- the object of interest is represented as a **single dot**
- all **external forces** on that object are drawn as arrows from the dot
- internal forces are excluded

Drawing an FBD

1. Draw a dot representing the object.
2. Identify **every** external force.
3. Draw each force as an arrow from the dot.
4. Label each arrow.



Normal Force and Weight

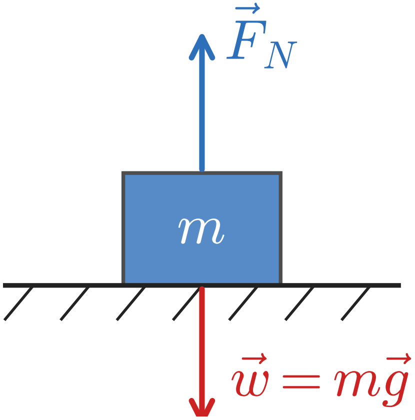
Normal Force $\left(\vec{F}_N\right)$ — SI Unit: N

The **normal force** is the contact force a surface exerts on an object, directed **perpendicular** to the surface.

- "Normal" means perpendicular (geometry term)
- Purpose: prevent objects from passing through each other
- On a flat, horizontal surface: $\vec{F}_N = -\vec{w} = m\vec{g}$ *(when $a_y = 0$)*

$\vec{F}_N \neq m\vec{g}$ in general. This holds **only** on a flat surface with no vertical acceleration.

$$\vec{a}_y = 0 \because \Sigma\vec{F}_y = 0$$



Tension and Friction

Tension $\left(\vec{T}\right)$ — SI Unit: N

Tension is the pulling force transmitted along a flexible connector (rope, cable, string).

- Tension always **pulls** — it cannot push.
- Acts along the connector, away from the object.

Friction $\left(\vec{f}\right)$ — SI Unit: N

Friction is a contact force between surfaces that opposes relative sliding motion.

- Always acts **opposite to the direction of motion** (or attempted motion)
- Friction is a real external force

You cannot push a rope.

Part III

Solving Newton's Second Law Problems

Outline of Part III: Solving Newton's Second Law Problems

Problem-Solving Strategy

Inclined Planes

Connected Systems and Tension

How to Solve a Dynamics Problem

Four-Step Strategy

1. **Draw a physical representation** of the situation.
2. **Draw a free-body diagram** (FBD) for each object of interest.
3. **Apply Newton's Second Law** in all relevant directions:

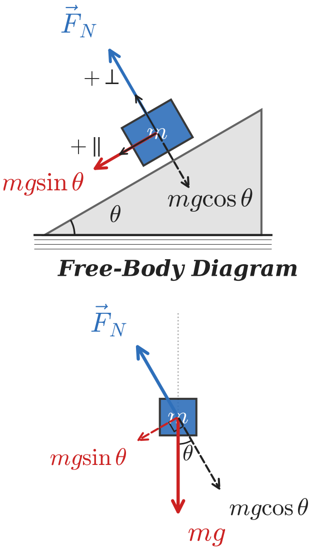
$$\sum \vec{F}_x = m\vec{a}_x \qquad \sum \vec{F}_y = m\vec{a}_y$$

   (or use $\parallel$ and $\perp$ axes when working on an inclined plane)
4. **Solve** the resulting equations for the unknown(s).

Check your answer: do the units work out? Is the magnitude reasonable? Does the direction make sense?

Outline of Part III: Solving Newton's Second Law Problems

Forces on an Inclined Plane



On an inclined plane at angle θ, it is convenient to choose axes **parallel** ($\parallel$) and **perpendicular** ($\perp$) to the slope.

Weight $\vec{w} = m\vec{g}$ has two components:

$$W_\parallel = mg\sin\theta \quad \text{(along slope)}$$

$$W_\perp = mg\cos\theta \quad \text{(into surface)}$$

The normal force equals the perpendicular component of weight (when $a_\perp = 0$):

$$\vec{F}_N = mg\cos\theta$$

Acceleration on a Frictionless Inclined Plane

Applying $\sum \vec{F}_{||} = m\vec{a}_{||}$ along the slope (taking down-slope as positive):

$$W_{||} = m\vec{a}_{||}$$

$$mg\sin\theta = m\vec{a}_{||}$$

$$\vec{a}_{||} = g\sin\theta$$

where $\vec{a}_{||}$ is the acceleration along the slope, $g = 9.80\,\frac{\text{m}}{\text{s}^2}$, and θ is the angle of the slope from horizontal.

Mass **cancels out** — all objects accelerate at the same rate down a frictionless incline regardless of mass.

Example: Snowboarder on a Slope

Anna Snowboards Down a Slope — (cp2e_ch4)

Anna ($m = 60.0\,\text{kg}$) snowboards down a slope inclined at $\theta = 10.0°$.
Assume the slope is frictionless.
Find (a) her acceleration down the slope and (b) the normal force.

$$m = 60.0\,\text{kg},\ \ \theta = 10.0°,\ \ g = 9.80\,\tfrac{\text{m}}{\text{s}^2}$$

$$\text{(a)} \quad \sum \vec{F}_{\parallel} = m\vec{a}_{\parallel}$$

$$mg\sin\theta = m\vec{a}_{\parallel}$$

$$\boxed{\vec{a}_{\parallel} = g\sin 10.0° = 1.70\,\tfrac{\text{m}}{\text{s}^2}}$$

Note: we used **two independent equations**, one for each direction.

The $\parallel$ equation gives $\vec{a}$; the $\perp$ equation gives $\vec{F}_N$.

- Mass cancels in part (a) — same $\vec{a}$ for any mass.
- $\vec{F}_N < mg$ on a slope ($\cos\theta < 1$).

Example: Snowboarder on a Slope

Anna Snowboards Down a Slope — (cp2e_ch4)

Anna ($m = 60.0\,\text{kg}$) snowboards down a slope inclined at $\theta = 10.0°$.
Assume the slope is frictionless.
Find (a) her acceleration down the slope and (b) the normal force.

$$m = 60.0\,\text{kg},\ \theta = 10.0°,\ g = 9.80\,\frac{\text{m}}{\text{s}^2}$$

$$\text{(b)} \quad \sum \vec{F}_\perp = 0 \quad \because \quad a_\perp = 0$$

$$\vec{F}_N - mg\cos\theta = 0$$

$$\boxed{\vec{F}_N = mg\cos 10.0° = 578\,\text{N}}$$

Note: we used **two independent equations**, one for each direction.

The $\parallel$ equation gives $\vec{a}$; the $\perp$ equation gives $\vec{F}_N$.

- Mass cancels in part (a) — same $\vec{a}$ for any mass.
- $\vec{F}_N < mg$ on a slope ($\cos\theta < 1$).

Check Your Understanding

A block slides down a frictionless ramp inclined at $25.0°$.

(a) What is its acceleration?
(b) Does the answer depend on the block's mass?

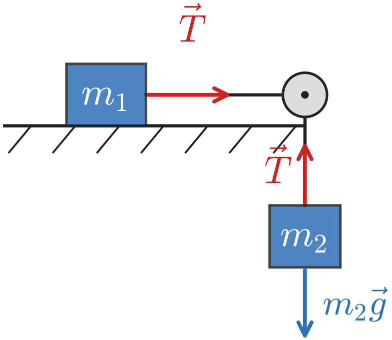
Come to lecture to see the answer.

Outline of Part III: Solving Newton's Second Law Problems

Objects Connected by a Rope

When two objects are connected by a **massless, inextensible** rope over a **frictionless pulley**:

- The tension T is the **same throughout** the rope.
- Both objects have the **same magnitude of acceleration**.



Strategy: draw a separate FBD for each object. Write $\sum \vec{F} = m\vec{a}$ for each. Use the shared T and a to link the equations.

- $T_1 = T_2 = T$ (massless rope)
- $a_1 = a_2 = a$ (inextensible rope)

Example: Modified Atwood Machine

Block on a Surface Connected to a Hanging Mass — (cp2e_ch4)

A 4.00 kg block sits on a frictionless horizontal surface, connected by a rope over a pulley to a hanging 3.00 kg mass.
Find (a) the acceleration of the system and (b) the tension in the rope.

$$m_1 = 4.00\,\text{kg},\ m_2 = 3.00\,\text{kg}$$

$$\text{For } m_1\ (x):\ T = m_1 a$$

$$\text{For } m_2\ (y):\ m_2 g - T = m_2 a$$

$$\text{Add equations:}\quad m_2 g = (m_1 + m_2) a$$

$$a = \frac{m_2 g}{m_1 + m_2} = \frac{(3.00)(9.80)}{7.00}$$

$$\boxed{a = 4.20\,\frac{\text{m}}{\text{s}^2}}$$

$$\boxed{T = m_1 a = (4.00)(4.20) = 16.8\,\text{N}}$$

Check:
$T = m_2(g - a) = 3.00(9.80 - 4.20) = 3.00(5.60) = 16.8\,\text{N}$ ✓

- Treat each mass separately.
- $T < m_2 g$ because m_2 is **accelerating** — not in equilibrium.

Part IV

Further Applications

Outline of Part IV: Further Applications

Drag Force and Terminal Velocity

The Four Basic Forces

Drag Force

Drag Force $\left(\vec{F}_D\right)$ — SI Unit: N

The **drag force** is the frictional force exerted by a **fluid** (air or water) on an object moving through it. It opposes the direction of motion:

$$F_D = \frac{1}{2} C \rho A v^2$$

where C is the dimensionless drag coefficient, ρ is the fluid density $\left(\frac{\mathrm{kg}}{\mathrm{m}^3}\right)$, A is the cross-sectional area $\left(\mathrm{m}^2\right)$, and v is the speed relative to the fluid.

- For air at 15°C: $\rho \approx 1.225\,\frac{\mathrm{kg}}{\mathrm{m}^3}$
- C is determined by experiment. This will be given to you for a problem.
- $F_D \propto v^2$ — drag grows rapidly with speed

Terminal Velocity

Terminal Velocity (v_e)

As an object falls and speeds up, drag increases. **Terminal velocity** is reached when drag equals weight and acceleration becomes zero:

$$\vec{F}_D + \vec{w} = 0 \quad \Rightarrow \quad F_D = w$$

$$\tfrac{1}{2} C\rho A v_e^2 = mg$$

$$v_e = \sqrt{\frac{2mg}{C\rho A}}$$

where v_e is the terminal (equilibrium) velocity and all other variables are as defined above.

At terminal velocity: $a_y = 0$ and $v =$ constant.
The object is in **dynamic equilibrium** in the vertical direction.

Example: Terminal Velocity of a Falling Squirrel

Terminal Velocity — (cp2e_ch4)

A squirrel falls from a tree. Its body acts like a parachute with dimensions $L = 30\,\mathrm{cm}$, $W = 25\,\mathrm{cm}$. Its mass is $m = 400\,\mathrm{g}$. Take $C = 1$, $\rho_{\mathrm{air}} = 1.225\,\frac{\mathrm{kg}}{\mathrm{m}^3}$. Find the terminal velocity.

$$A = LW = (0.30\,\mathrm{m})(0.25\,\mathrm{m}) = 0.075\,\mathrm{m}^2$$

$$m = 0.400\,\mathrm{kg}$$

$$v_e = \sqrt{\frac{2mg}{C\rho A}} = \sqrt{\frac{2(0.400)(9.80)}{(1)(1.225)(0.075)}}$$

$$v_e = \sqrt{\frac{7.84}{0.0919}} = \sqrt{85.3}$$

$$\boxed{v_e = 9.24\,\frac{\mathrm{m}}{\mathrm{s}} \approx 9.2\,\frac{\mathrm{m}}{\mathrm{s}}}$$

$9.2\,\frac{\mathrm{m}}{\mathrm{s}} \approx 21\,\frac{\mathrm{m}}{\mathrm{h}}$

Squirrels are well-designed for this: large surface area relative to body mass lowers terminal velocity. They survive such falls routinely.

- Larger $A \Rightarrow$ lower v_e (parachute effect)
- Larger $m \Rightarrow$ higher v_e

Outline of Part IV: Further Applications

The Four Fundamental Forces of Nature

All forces in nature reduce to exactly **four fundamental forces**:

Force	**Rel. Strength**	**Range**	**Everyday Role**
Strong nuclear	1	$\sim 10^{-15}$ m	Holds nucleus together
Electromagnetic	10^{-2}	Infinite	Contact, friction, tension
Weak nuclear	10^{-6}	$< 10^{-18}$ m	Radioactive decay
Gravitational	10^{-38}	Infinite	Weight, orbits

- Nearly all everyday forces (contact, friction, tension, normal) are **electromagnetic** at the atomic level.
- Gravity is extraordinarily weak but dominates at astronomical scales because it is always attractive and has infinite range.
- Electromagnetic + weak nuclear = **electroweak force** (unified at high energy).

Next Lecture

Chapter 5: Further Applications of Newton's Laws

- Static and kinetic friction — $f_k = \mu_k F_N$
- Drag force in more detail; Stokes' law
- Hooke's Law and elasticity: $\vec{F}_s = -k\Delta\vec{x}$
- Young's modulus and stress/strain

- The **four-step strategy** from today applies to all problems in Chapter 5.
- Review today's examples — inclined planes and connected systems appear frequently.